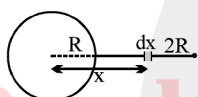


HINTS AND SOLUTIONS

- Q.1 A
Q.2 A
Q.3 C
Q.4 B
Q.5 B
Q.6 C
Q.7 B
Q.8 A
Q.9 C
Q.10 D
Q.11 D
Q.12 C
Q.13 B
Q.14 A
Q.15 B

Sol. Gravitation/SCQ/93



Force on this element

$$dF = \frac{GM(\lambda dx)}{x^2}$$

∴ Total force on the stick

$$\int_0^F dF = GM\lambda \int_R^{3R} \frac{dx}{x^2}$$

$$F = GM\lambda \left[\frac{-1}{x} \right]_R^{3R} = GM\lambda \left[\frac{1}{R} - \frac{1}{3R} \right] = GM\lambda \left[\frac{2}{3R} \right]$$

CM of stick is rotating in a circle of radius 2R

$$\therefore GM \left[\frac{2\lambda}{3R} \right] = \lambda(2R)(2R)\omega^2$$

$$\omega^2 = \frac{GM}{6R^3}$$

$$T = 2\pi\sqrt{6} \sqrt{\frac{R^3}{gR^2}} = 2\pi\sqrt{6} \sqrt{\frac{R}{g}}$$

Q.16 AC

Q.17 D

Sol. Heat produced = $W_F - \text{E.P.E.}$

$$= F\ell - \frac{F\Delta}{2} = \frac{F\Delta}{2}.$$

Q.18 AD

Sol. $\Delta\ell = \frac{F\Delta_0}{A_y}$

$$\Rightarrow \frac{\Delta_0}{A} = \frac{\Delta_0}{A'}$$

$$\frac{0.6}{15} = \frac{0.4}{A'} \Rightarrow A' = 10 \text{ cm}^2$$

$$\text{stress} = \frac{F}{A} \Rightarrow \frac{\sigma_1}{\sigma_2} = \frac{A_2}{A_1} = \frac{10}{15}$$

$$\frac{\Delta r}{r} = -\sigma \frac{\Delta\ell}{\ell}$$

$$\Delta r = -\sigma \frac{\Delta\ell}{\ell} \times r = -\sigma \Delta\ell \frac{\sqrt{A}}{\ell}$$

$$\Delta r_{\text{left}} = \frac{\sqrt{15}}{0.6} = \sqrt{\frac{1500}{0.36}}$$

$$\Delta r_{\text{right}} = \frac{\sqrt{10}}{0.4} = \sqrt{\frac{1000}{0.16}}$$

Q.19 ABC

Sol. $2T(\Delta\theta/2) = N \cos 45^\circ$

$$\Rightarrow T = mg/2\pi$$

$$\Delta l = \frac{T(2\pi R)}{aY}$$

Q.20 ACD

Q.21 [Ans. (A) Q, R; (B) R; (C) P; (D) P, S]

Sol.(A) For $x < x_1$ graph is straight line means it is proportionality limit means Hooke's law is valid $\rightarrow$ Q

Permanent deformation takes place after B elastic limit $\rightarrow$ R

(B) $x_1 < x < x_2$ between elastic limit & proportionality limit $\rightarrow$ R

(C) P

(D) $x > x_3$ is beyond Breaking point $\rightarrow$ P, S

Q.22 [Ans. $E = \frac{t \pi \rho^2 \omega^4 r^6}{6Y}$]

Q.23

[Sol. Gravitation/OMR/5 Form energy conservation

$$\frac{1}{2} m v_0^2 + \frac{1}{2} m v^2 = 2 \left(\frac{1}{2} m v^2 \right) - \frac{Gm}{r} \quad \dots (1)$$

from angular momentum conservation about the centre of any one body

$$m v_0 L = m v r$$

$$v = \frac{v_0 L}{r} \quad \dots (2)$$

solve equation (1) & (2) & get $r = \frac{L + \sqrt{5}L}{2}$

Q.24 [Ans. $v_0^2 - v^2 = \frac{(2gh)}{\left(1 + \frac{h}{r}\right)}, \frac{81}{19}R$]

Q.25 [Ans. $\Delta r = \frac{1.25}{\pi} \times 10^{-6} \text{ m}$]

Sol. $\sigma = \frac{\left(\frac{\Delta r}{r}\right)}{\left(\frac{\Delta l}{l}\right)}$

$$\sigma \times \frac{\Delta l}{l} = \frac{\Delta r}{r}$$

$$\Delta r = \sigma \left(\frac{\Delta l}{l} \right) r$$

$$\Delta r = \left[\sigma \frac{F}{AY} r \right]$$

$$\Delta r = 0.25 \times \frac{100 \times 10^{-3}}{\pi \times (10^{-3}) \times 2 \times 10^{10}}$$

$$\Delta y = \frac{25 \times 1}{\pi \times 10^{-3} \times 2 \times 10^{10}}$$

$$\Delta r = \frac{25}{\pi \times 2 \times 10^7} = \left[\frac{25}{20\pi} \times 10^{-6} \right]$$

$$\Delta r = \left[\frac{1.25}{\pi} \times 10^{-6} \right] \quad \text{Ans.}$$